Chapter 9: Statics and Torque

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Lecture 9

Part I

Static Equilibrium

Outline of Part I: Static Equilibrium

What is Statics?

Equilibrium: $\vec{F}_{\text{net}} = m\vec{a} = 0$ (linear) $\quad \vec{\tau}_{\text{net}} = I\vec{\alpha} = 0$ (rotational)

An object is in **equilibrium** whenever the net force *and* the net torque are simultaneously zero ($\vec{a} = 0$).

Static Equilibrium

- $\vec{v} = 0$ (at rest)
- $\vec{a} = 0$ (remains at rest)

Dynamic Equilibrium

- $\vec{v} \neq 0$ (in motion)
- $\vec{a} = 0$ (remains in motion)

The study of **statics** is on static equilibrium. Many structures we used must remain still and must not accelerate linearly *or rotationally*.

Two Conditions for Static Equilibrium

First Condition

No translational acceleration:

$$\vec{F}_{\text{net}} = 0 \quad \Rightarrow \quad \begin{cases} \sum \vec{F}_x = 0 \\ \sum \vec{F}_y = 0 \end{cases}$$

Second Condition

No rotational acceleration:

$$\vec{\tau}_{\text{net}} = 0 \quad \Rightarrow \quad \sum \vec{\tau} = 0$$

Both conditions must hold *simultaneously* for static equilibrium.

An object can satisfy the first condition yet still rotate (e.g., a couple of equal and opposite forces not on the same line of action).

Outline of Part I: Static Equilibrium

First Condition for Equilibrium

First Condition: $\sum \vec{F} = 0$

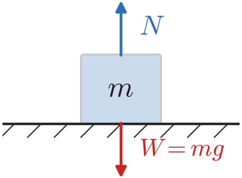
The vector sum of all external forces acting on an object must be zero:

$$\sum \vec{F}_x = 0 \qquad \text{and} \qquad \sum \vec{F}_y = 0$$

- This is Newton's Second Law applied to both directions simultaneously.
- It prevents the object from *accelerating* in any direction.
- It does *not* prevent the object from rotating.

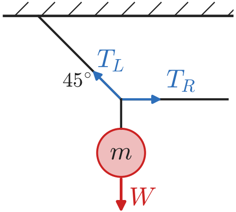
Block on Surface

$\sum \vec{F}_y = N - W = 0$



Two-Rope Hang

$\sum \vec{F}_x = 0, \ \sum \vec{F}_y = 0$



Strategy for Satisfying 1EQ:

Draw all forces on a free-body diagram, then resolve into components and set each sum to zero.

Example: First Condition

A traffic light of weight $W = 980\,\text{N}$ is supported by two cables. The left cable makes 45.0° with the horizontal; the right cable is horizontal. Find the tension in each cable.

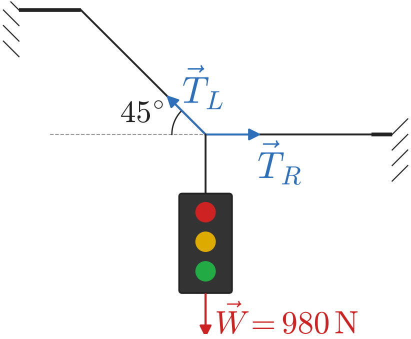
$$\sum \vec{F}_y = 0: \quad T_L \sin 45^\circ - W = 0$$

$$T_L = \frac{W}{\sin 45^\circ} = \frac{980}{0.707}$$

$$\boxed{T_L = 1386\,\text{N} \approx 1390\,\text{N}}$$

$$\sum \vec{F}_x = 0: \quad T_R - T_L \cos 45^\circ = 0$$

$$\boxed{T_R = T_L \cos 45^\circ = 980\,\text{N}}$$



The angled cable $\vec{T}_L$ supports all the vertical load (as $\vec{T}_{Ly}$) and provides a horizontal component ($\vec{T}_{Lx}$) that the horizontal cable $\vec{T}_R$ must balance.

Notice: $|\vec{T}_L| > |\vec{W}|$, which you may find surprising.

Check Your Understanding

A picture frame of weight 50.0 N hangs from two wires that each make an angle of 30.0° with the horizontal.

What is the tension in each wire?

Come to lecture to see the answer.

Outline of Part I: Static Equilibrium

Torque: The Rotational Analogue of Force

Just as **force** causes linear acceleration, **torque** causes rotational (angular) acceleration.

Torque is the tendency of a force to rotate an object about a pivot point.

- Pushing a door near the hinges requires more force than pushing near the edge.
- A longer wrench makes loosening a bolt easier.
- These are everyday examples of torque — the same force applied farther from the pivot produces more torque.

torquēre [ˈtor.kʷeː.re] *verb* – Latin

to twist, to wrench

Descendants: **torque**, "torture", "contort", "distort"

Torque: Definition and Formula

Torque (τ) — SI Unit: newton·meter $(\mathrm{N \cdot m})$

The torque produced by a force $\vec{F}$ applied at position $\vec{r}$ from the pivot is:

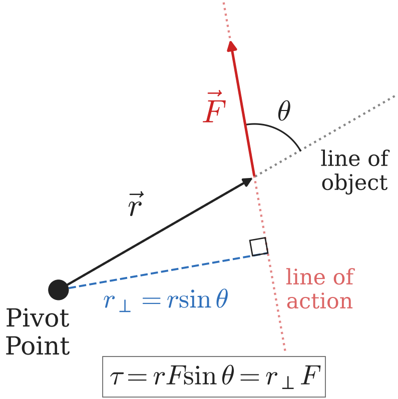
$$\tau = rF\sin\theta_{\vec{r}\vec{F}}$$

where r is the **distance from the pivot to where the force is applied**, F is the magnitude of the force, and $\theta_{\vec{r}\vec{F}}$ is the angle between $\vec{r}$ and $\vec{F}$.

Lever/Moment Arm, $r_\perp$

$$\tau = \underbrace{(r\sin\theta_{\vec{r}\vec{F}})}_{r_\perp} F = r_\perp F$$

$r_\perp$ is measured *perpendicular to the force*, from the pivot point to where $\vec{F}$ is applied.



- Max torque when $\theta = 90°$: $\tau_{\max} = rF$.
- Zero torque when $\theta = 0°$ or $\theta = 180°$ (force along $\vec{r}$).

Sign Convention for the Torque Vector $\vec{\tau}$

Counterclockwise (CCW) = Positive $\vec{\tau}$

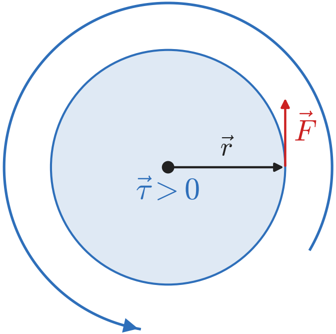
If the force by itself would cause **counterclockwise** rotation, it contributes **positive** torque.

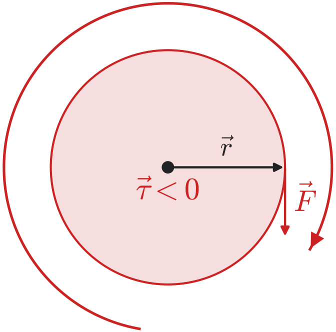
$$\vec{\tau} > 0$$

Clockwise (CW) = Negative $\vec{\tau}$

If the force by itself would cause **clockwise** rotation, it contributes **negative** torque.

$$\vec{\tau} < 0$$





Example: Calculating Torque

A force of 25.0 N is applied to a wrench at a distance of 0.300 m from the bolt.

(a) Find the torque if the force is applied perpendicular to the wrench.

(b) Find the torque if the force is applied at 60.0° to the wrench.

$$\text{(a)}\quad \tau = rF\sin 90^\circ = (0.300\,\mathrm{m})(25.0\,\mathrm{N})(1)$$

$$\boxed{\tau = 7.50\,\mathrm{N}\cdot\mathrm{m}}$$

$$\text{(b)}\quad \tau = rF\sin 60^\circ = (0.300\,\mathrm{m})(25.0\,\mathrm{N})(0.866)$$

$$\boxed{\tau = 6.50\,\mathrm{N}\cdot\mathrm{m}}$$

Applying the force perpendicular to the wrench maximizes torque. Angling the force reduces its effectiveness.

Check Your Understanding

A force of 40.0 N is applied at the end of a 0.500 m rod, perpendicular to the rod.

(a) What is the torque about the pivot at the other end of the rod?

(b) If the force is instead applied parallel to the rod, what is the torque?

Come to lecture to see the answer.

Check Your Understanding

You are trying to loosen a stuck bolt with a wrench.

Describe how you would increase the torque applied to the bolt without increasing the magnitude of the force.

Come to lecture to see the answer.

Outline of Part I: Static Equilibrium

Second Condition for Equilibrium

Second Condition: $\sum \vec{\tau} = 0$

For an object in rotational equilibrium, the net torque about *any* pivot point must be zero:

$$\sum \vec{\tau} = 0$$

Equivalently: the sum of all CCW torques equals the sum of all CW torques.

- This ensures the object has no angular acceleration ($\alpha = 0$).
- This condition must hold about **any** chosen pivot point (not just a special one).
- Choosing the pivot at an unknown force's point of application eliminates that force from the torque equation.

Choosing the Pivot Point Strategically

Any force that acts **at the pivot point** produces **zero torque** about that pivot ($\because r = 0$).

$\therefore$ If there is a force with unknown magnitude, place the pivot at the point where the unknown force acts, since it is not considered in the torque equation.

Use This to Solve Problems More Easily

- If you have two unknown forces, place the pivot at one of them.
- The torque equation will then contain only *one* unknown.
- Solve the torque equation, then use $\sum \vec{F} = 0$ to find the second unknown.

The physics does not change with your choice of pivot — equilibrium holds about any point. You are free to choose whichever makes the algebra simplest.

Example: Seesaw — Find the Unknown Mass

Alice ($m_A = 40.0\,\mathrm{kg}$) and Billy sit on opposite sides of a seesaw. Alice sits $r_A = 1.00\,\mathrm{m}$ from the fulcrum; Billy sits $r_B = 1.50\,\mathrm{m}$ from the fulcrum. Find Billy's mass m_B so the seesaw is in static equilibrium.

1. Pivot at fulcrum:

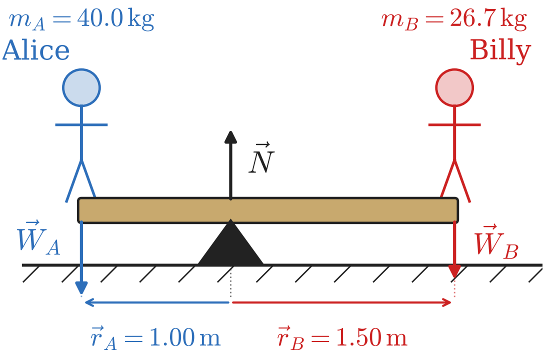
$$\sum \vec{F}_y = 0 : \quad N = (m_A + m_B)g$$

$$\sum \vec{\tau} = 0 : \quad \vec{\tau}_A + \vec{\tau}_B + \cancel{\vec{\tau}_N} = 0$$

$$r_A m_A g - r_B m_B g = 0$$

$$r_A m_A = r_B m_B$$

$$m_B = m_A\left(\frac{r_A}{r_B}\right) = 40.0\left(\frac{1.00}{1.50}\right) = \boxed{26.7\,\mathrm{kg}}$$



Why does $\tau_N = 0$?

Answer is Independent of Pivot Choice

Equilibrium holds about *any* pivot, so we should be able to verify the seesaw result by choosing a *different* pivot point.

2. Pivot at Alice's seat (left end):

- $\vec{\tau}_A = 0 \quad (\because r_A = 0 \text{ at the pivot})$
- $\vec{N}$ acts 1.00 m right $\Rightarrow$ CCW (+)
- $\vec{W}_B$ acts 2.50 m right $\Rightarrow$ CW (−)

$$\sum \vec{F}_y = 0: \quad N = (m_A + m_B)g$$

$$\sum \vec{\tau} = 0: \quad \cancel{\vec{\tau}_A} + \vec{\tau}_B + \vec{\tau}_N = 0$$

$$-r_B m_B g + r_N (m_A + m_B) g = 0$$

$$m_B = \frac{r_N m_A}{r_B - r_N} = \frac{(1)(40)}{2.5 - 1} = \boxed{26.7\,\text{kg}}$$

Freedom in Pivot Choice

Pivot choice is not a physical constraint and ∴ does not affect the answer.

You should therefore place the pivot where an *unknown* force acts so it contributes zero torque (given your pivot choice, since $r = 0$ in that case) and drops out of the equation.

This saves you the trouble of solving for the unknown force.

What was the **unknown force** here?

Check Your Understanding

Two children sit on a seesaw pivoted at its center. Child A (35.0 kg) sits 1.20 m from the center. How far from the center must Child B (42.0 kg) sit to balance the seesaw?

Come to lecture to see the answer.

Example: Person on a Plank

A uniform plank ($m_p = 30.0\,\text{kg}$, $L = 4.00\,\text{m}$) is supported at both ends (left support F_L, right support F_R). Stuart ($m_s = 70.0\,\text{kg}$) stands 1.00 m from the right end. Find F_L and F_R.

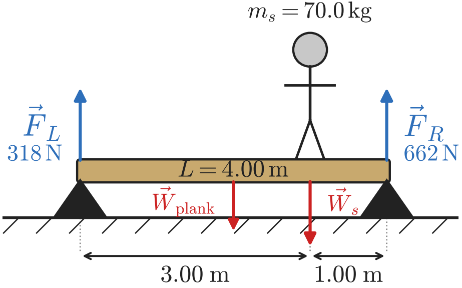
Pivot at left end (eliminates $\vec{F}_L \because \vec{r}_L = 0$):

$$\sum \vec{\tau} = 0 : \quad \cancel{\vec{\tau}_L} + \vec{\tau}_p + \vec{\tau}_s + \vec{\tau}_R = 0$$

$$-r_p W_p - r_s W_s + r_R F_R = 0$$

$$4.00\,F_R = 2058 + 588 = 2646$$

$$\boxed{F_R = 662\,\text{N}}$$



$$\sum \vec{F}_y = 0 : \; F_L + F_R = (70 + 30)(9.8) = 980\,\text{N}$$

$$\boxed{F_L = 980 - 662 = 318\,\text{N}}$$

Stuart is closer to the right support, so $F_R > F_L$.

What kind of force is F_L and F_R?

Check Your Understanding

A uniform 4.00 m seesaw has mass 20.0 kg. Child A (30.0 kg) sits at the left end; Child B (40.0 kg) sits at the right end. The seesaw pivots at its center.

Is the seesaw in rotational equilibrium? If not, which way does it rotate?

Come to lecture to see the answer.

Check Your Understanding

Where must Child B (40.0 kg) sit from the center of the seesaw to balance Child A (30.0 kg) sitting 2.00 m from the center?

Come to lecture to see the answer.

Check Your Understanding

Can an object satisfy the first condition for equilibrium ($\sum \vec{F} = 0$) but *not* the second ($\sum \vec{\tau} = 0$)?

Give an example.

Come to lecture to see the answer.

Part II

Applications of Statics

Outline of Part II: Applications of Statics

Problem-Solving Strategy for Statics

1. **Draw a free-body diagram.** Show every external force acting on the object with arrows indicating direction.
2. **Choose a coordinate system.** The conventional $+x$ right, $+y$ up is usually best.
3. **Identify all forces** and resolve them into x- and y-components.
4. **Choose a pivot point.** Place it where an unknown force acts to simplify the torque equation.
5. **Write the three equilibrium equations:**
   $$\sum \vec{F}_x = 0, \quad \sum \vec{F}_y = 0, \quad \sum \vec{\tau} = 0$$
6. **Solve** the system of equations for the unknowns.
7. **Check** your answers for physical reasonableness.

Example: Beam Leaning Against a Wall

A uniform beam ($m = 40.0\,\mathrm{kg}$, $L = 5.00\,\mathrm{m}$) leans against a frictionless wall at $\theta = 60.0^\circ$ from the floor. Find the normal force from the wall (N_w) and the reaction at the floor.

Pivot at floor contact

(eliminates torque due to floor forces):

$\sum \vec{F}_x = 0$: $F_{\mathrm{floor},x} = N_w = 113\,\mathrm{N}$

$\sum \vec{F}_y = 0$: $F_{\mathrm{floor},y} = mg = 392\,\mathrm{N}$

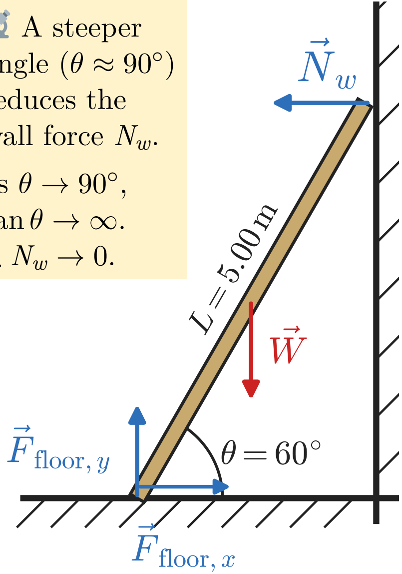
$$\sum \vec{\tau}_{\mathrm{floor}} = 0: \quad N_w(L\sin\theta) = mg\left(\tfrac{L}{2}\cos\theta\right)$$

$$N_w = \frac{mg\cos\theta}{2\sin\theta} = \frac{mg}{2\tan\theta}$$

$$\boxed{N_w = \frac{(40.0)(9.80)}{2(1.732)} = 113\,\mathrm{N}}$$

A steeper angle ($\theta \approx 90^\circ$) reduces the wall force N_w.

as $\theta \to 90^\circ$,
$\tan\theta \to \infty$.
$\therefore N_w \to 0$.



Check Your Understanding

In the beam-against-wall problem, what happens to the force the wall must exert as the angle θ (measured from the floor) increases toward $90°$?

What happens as θ decreases toward $0°$?

Come to lecture to see the answer.

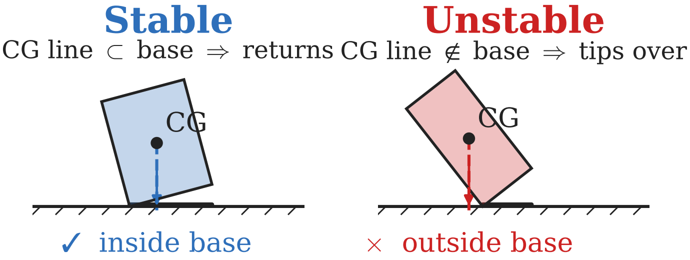
Outline of Part II: Applications of Statics

Center of Gravity and Stability

Center of Gravity (CG)

The **center of gravity** is the point at which the weight of an object can be considered to act.

For uniform gravity, CG coincides with the **center of mass**.



Stability Rule

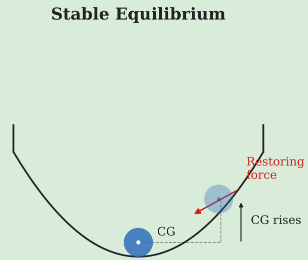
An object is in **stable** equilibrium as long as the vertical line through its CG falls *within* its base of support.

If it falls *outside* the base, the object tips over.

- **Lower CG** ⇒ more stable
- **Wider Support Base** ⇒ more stable

Types of Equilibrium (CG: Center of Gravity)

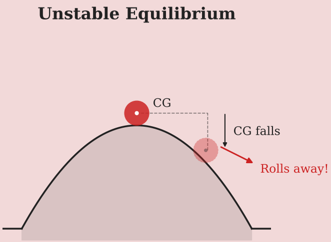
Stable



Displaced ⇒ returns to equilibrium.

CG *rises* when displaced.

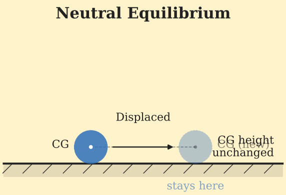
Unstable



Displaced ⇒ moves *away* from equilibrium.

CG *falls* when displaced.

Neutral



Displaced ⇒ stays in new position.

CG height *unchanged*.

Check Your Understanding

A race car is designed with a very low center of gravity and wide wheelbase. A bus, by contrast, has a high center of gravity and a relatively narrow wheelbase.

Explain why race cars rarely tip over in corners, while loaded buses and trucks occasionally do.

Come to lecture to see the answer.

Check Your Understanding

A person leans forward while carrying a heavy backpack. Why do they lean forward, and why would leaning backward cause them to fall?

Come to lecture to see the answer.

Outline of Part II: Applications of Statics

Problem-Solving Strategy

Stability

Simple Machines

Forces and Torques in Muscles and Joints

Simple Machines and (Ideal) Mechanical Advantage

Mechanical Advantage (MA)

A **simple machine** exploits torque balance to allow differences in **input force (effort)** and **output force (load)**:

$$MA = \frac{F_{\text{out}}}{F_{\text{in}}} = \frac{d_{\text{in}}}{d_{\text{out}}}$$

where d_{in} and d_{out} are the distances (or lever arms) for the input and output forces.

$MA > 1$: machine amplifies **force** (at the cost of distance traveled).
$MA < 1$: machine amplifies **distance/speed** (at the cost of force).

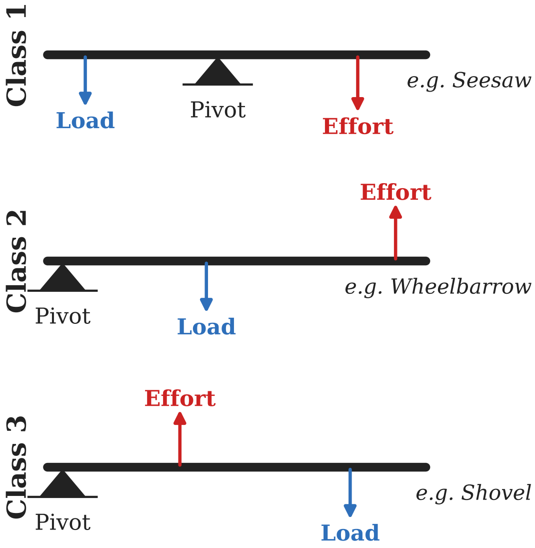
🎗 **Real** machines: $MA_{\text{actual}} < MA_{\text{ideal}}$ due to friction.

Common Simple Machines

- **Lever (3 classes)**, **Wheel and Axle**, **Pulley**
- **Inclined Plane**, **Wedge**, **Screw**

Note: the **load** is also known as the **resistive force**.

Classes of Levers



Third-class levers ($MA < 1$) trade force for *speed* and range of motion.

The forearm is a 3rd-class lever — muscles must exert forces much larger than the load.

A shovel is also a 3rd-class lever — it's useful because dirt weighs little and we move a lot of it.

Classes of Levers: Summary Table

Class	Pivot location	MA	Examples
1st	Between effort and load	> 1, $= 1$, or < 1	Seesaw, scissors, pliers
2nd	Load between pivot and effort	Always > 1	Wheelbarrow, nutcracker
3rd	Effort between pivot and load	Always < 1	Tweezers, forearm, broom

Check Your Understanding

A lever has an input arm of 1.20 m and an output arm of 0.30 m.

(a) What is its mechanical advantage?

(b) If the output force needed is 400 N, what input force is required?

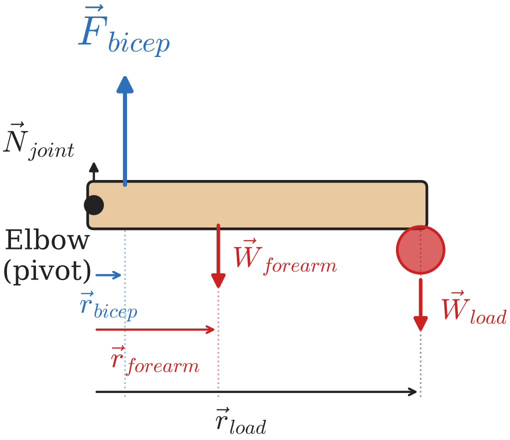
Come to lecture to see the answer.

Outline of Part II: Applications of Statics

Torques in the Human Body

The human body is a system of levers. Bones are the levers, joints are the pivots, and muscles provide the forces.

- Most muscles attach close to the joint (short lever arm for the muscle).
- The load (weight of limb + held object) acts farther from the joint.
- $\therefore$ muscles must exert forces *much larger* than the weight being supported.
- This design favors **speed and range of motion** over force.



📝 The forearm is a **third-class lever**: effort (muscle) is between the pivot (joint) and the load $\Rightarrow MA < 1$.

Example: Forearm as a Lever

A person holds a 10.0 kg ball in their hand. The forearm (mass 2.50 kg) is horizontal. The bicep attaches 4.00 cm from the elbow. The forearm's CG is 16.0 cm from the elbow, and the ball is 38.0 cm from the elbow. Find the bicep force F_B.

Pivot at elbow joint (eliminates joint force):

$$\sum \vec{\tau}_{\text{elbow}} = 0 : \quad F_B(0.0400) - (2.50)(9.80)(0.160) - (10.0)(9.80)(0.380) = 0$$

$$F_B(0.0400) = 3.92 + 37.24 = 41.16$$

$$\boxed{F_B = 1029\,\text{N} \approx 1030\,\text{N}}$$

The bicep must exert 1030 N — over **ten times** the weight of the ball!

$$MA = \frac{F_{\text{out}}}{F_{\text{in}}} = \frac{(10.0)(9.80)}{1030} \approx 0.095$$

As expected for a 3rd-class lever.

Check Your Understanding

A person stands on their tiptoes. Their calf muscle (Achilles tendon) pulls upward on the heel, and the ball of the foot pushes down on the floor.

Is the foot a first-, second-, or third-class lever in this scenario?

Does the calf muscle exert more or less force than the person's body weight? Explain.

Come to lecture to see the answer.

Check Your Understanding

Why is it harder to hold a heavy grocery bag with your arm fully extended than with your arm bent at the elbow?

Come to lecture to see the answer.

Chapter Summary

Key Equations

- $\sum \vec{F}_x = 0 \quad \text{(1st condition)}$
- $\sum \vec{F}_y = 0 \quad \text{(1st condition)}$
- $\sum \vec{\tau} = 0 \quad \text{(2nd condition)}$
- $\tau = rF \sin \theta_{\vec{r}\vec{F}}$
- $MA = \dfrac{F_{\text{out}}}{F_{\text{in}}} = \dfrac{d_{\text{in}}}{d_{\text{out}}}$

Key Concepts

- Both conditions must hold for static equilibrium.
- Torque depends on force, distance, *and* angle.
- Choose the pivot to eliminate unknown forces.
- Stability: CG must be above the base of support.
- Human body: muscles in 3rd-class lever arrangements exert forces much larger than the load.